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Operating system and system programming Individual Assignment

Assigned Operating System : Qubes

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A. Introduction

Background

Qubes OS is a highly security-oriented operating system that redefines how system protection is implemented at the architectural level. Traditional operating systems such as Windows 10 or Ubuntu rely on a monolithic structure where applications share the same operating environment. While these systems use permission controls and antivirus tools, they still operate within a single kernel space, which means a successful attack can potentially affect the entire system.

Qubes OS introduces an advanced security model known as **"**Security by Compartmentalization." Instead of relying solely on defensive software mechanisms, it enforces isolation at the system level. This is achieved through the use of the Xen Hypervisor, a Type 1 (bare-metal) hypervisor that enables the system to run multiple isolated environments simultaneously.

Each of these isolated environments, called Qubes (virtual machines), is dedicated to a specific purpose. For example, one Qube may be used for personal activities, another for work-related tasks, and another for handling untrusted files or browsing unknown websites. This strict separation ensures that if one Qube is compromised—for instance, by malware or a malicious file—the attack is contained within that Qube and cannot spread to others.

This layered isolation significantly reduces the system's attack surface and provides a level of security that is far superior to conventional operating systems. As a result, Qubes OS is widely recognized as one of the most secure desktop operating systems available, particularly for users who require strong protection against advanced cyber threats.

Motivation

The motivation behind this project is both academic and practical. From an academic perspective, the project fulfills the requirements of an Operating Systems and System Programming course, where students are expected to demonstrate both theoretical understanding and practical implementation skills related to system installation, configuration, and virtualization.

From a practical perspective, working with Qubes OS provides valuable exposure to modern security-focused system design. Unlike traditional operating systems, Qubes OS requires the user to think in terms of isolation, trust levels, and compartmentalization. This helps develop a deeper understanding of how operating systems can be structured to defend against real-world cyber threats.

Furthermore, installing Qubes OS within VMware introduces the concept of nested virtualization, where one virtualization layer operates inside another. This is a critical concept in modern computing environments, especially in cloud computing, cybersecurity labs, and enterprise systems. Gaining experience with nested virtualization strengthens technical skills and prepares students for advanced topics in system administration and infrastructure design.

B. Objectives

1. Successful Deployment

The primary objective of this project is to successfully install and operate Qubes OS within a VMware Workstation environment. This involves not only completing the installation process but also ensuring that the system runs stably and efficiently. Achieving this requires careful configuration of virtual hardware, including memory allocation, processor cores, and storage settings.

Additionally, compatibility between the host system and the guest operating system must be ensured. Since Qubes OS has strict hardware requirements, successfully deploying it in a virtualized environment demonstrates a strong understanding of system compatibility and virtualization constraints.

2. Nested Virtualization Mastery

Another key objective is to understand and implement nested virtualization, a concept where one hypervisor runs inside another. In this case, VMware Workstation (a Type 2 hypervisor) hosts Qubes OS, which itself uses the Xen hypervisor (Type 1).

This layered virtualization introduces additional complexity, as hardware virtualization features must be passed from the physical CPU to VMware and then to Qubes OS. Successfully configuring this setup demonstrates advanced technical skills and a clear understanding of how virtualization works at multiple levels.

3. Security Architecture Analysis

A major goal of this project is to analyze the internal security architecture of Qubes OS. This includes understanding how different domains function and interact:

- **AppVMs**: These are application environments used for daily tasks such as browsing or document editing. They are isolated from each other to prevent data leakage.

- **NetVM**: This domain is responsible for handling all network-related operations. By isolating networking, Qubes OS prevents direct internet access from application environments.
- **Dom0**: This is the administrative domain that controls the entire system. It has access to hardware but is intentionally isolated from network access to reduce security risks.

Understanding these components provides insight into how Qubes OS achieves its strong security guarantees through isolation and minimal trust.

4. Practical System Configuration Skills

This project also aims to develop hands-on skills in configuring and managing system resources. Running Qubes OS requires careful allocation of CPU, RAM, and disk space due to its heavy reliance on virtualization.

Students gain experience in:

- Adjusting virtual machine settings
- Managing system performance
- Troubleshooting installation and runtime issues

These practical skills are essential for real-world system administration and virtualization tasks.

C. Requirements

i. Hardware Requirements

- **Processor (CPU):** A modern 64-bit processor with virtualization support (Intel VT-x or AMD-V) is required. Additionally, Extended Page Tables (EPT) or Rapid Virtualization Indexing (RVI) are necessary to improve memory management performance in virtualized environments.
  This requirement is critical because Qubes OS itself relies heavily on virtualization, and running it inside VMware adds another layer of virtualization.
- **Memory (RAM):** While the minimum requirement is 6 GB, this is only sufficient for basic functionality. In practice, Qubes OS benefits greatly from higher memory capacity, with 16 GB being the recommended amount.
  This is because multiple Qubes (virtual machines) run simultaneously, each consuming memory resources.

- **Storage:** A minimum of 50 GB of storage is required, but more space is recommended for better usability. Using an SSD significantly improves performance, especially during system boot, VM initialization, and disk-intensive operations.

ii. Software Requirements

- **VMware Workstation:** This software acts as the host hypervisor, allowing users to create and manage virtual machines. It provides the environment in which Qubes OS will be installed and executed.
- **Qubes OS ISO (R4.x):** This is the installation image used to install the operating system. It contains all necessary system files and components required for setting up Qubes OS.
- **Host Operating System:** A stable and compatible operating system such as Windows 11 or a Linux distribution is required to run VMware Workstation. The host OS must support hardware virtualization and have sufficient system resources.

D. Installation Steps

I. Download Qubes OS (ISO File)

1. Open your browser
2. Go to the official Qubes OS website
3. Click **Download Qubes OS**
4. Choose the latest version
5. Click the **ISO file** to start downloading

II .Installing VMware Workstation Pro

Step 1: Download VMware

- ❖ Open a web browser and go to the official VMware website
- ❖ Click on **"Download Now"** to get VMware Workstation Pro

Step 2: Sign in to Broadcom

- ❖ Sign in to your Broadcom account (or create one if needed)
- ❖ After logging in, go to the **"My Downloads"** section

- ❖ Find VMware Workstation Pro from the available downloads

- ❖ Complete any required verification steps if prompted

Step 3: System Preparation and Optimization

A. Disable Conflicting Windows Features

To make sure VMware works properly, some Windows features need to be turned off:

- ❖ Open **"Turn Windows features on or off"** from the Control Panel

- ❖ Uncheck the following options:

- Hyper-V
- Virtual Machine Platform
- Windows Hypervisor Platform
- Windows Sandbox
- Windows Subsystem for Linux (WSL)

- ❖ Apply the changes and restart your computer

B. Security and BIOS Settings

- ❖ **Disable Memory Integrity:**

- Open Windows Security → Device Security
- Go to Core Isolation details
- Turn off Memory Integrity and restart if required

- ❖ **Check Virtualization Status:**

- Open Task Manager → Performance tab
- Make sure virtualization is enabled

- ❖ **If it is not enabled:**

- Restart your computer and enter BIOS/UEFI settings
- Enable:
    - Intel VT-x (for Intel processors)
    - AMD-V/SVM (for AMD processors)

- ❖ This step is important to run 64-bit virtual machines smoothly

Step 4: Install VMware Workstation Pro

- ❖ Run the downloaded installer file
- ❖ Follow the setup instructions:
  - Accept the license agreement
  - Make sure Hyper-V is disabled for better performance
  - Choose installation options like shortcuts if needed
- ❖ Click **"Install"** to start the installation
- ❖ Click **"Finish"** when the installation is complete

1. Virtual Machine Creation

- Open **VMware Workstation**
- Select Create New Virtual Machine (Typical)
- Choose the Qubes OS ISO file

Explanation:
This step initializes the virtual environment where Qubes OS will be installed. Selecting the "Typical" configuration simplifies the setup process by automatically applying standard settings. The ISO file acts as the installation media, similar to a bootable disk used in physical installations.

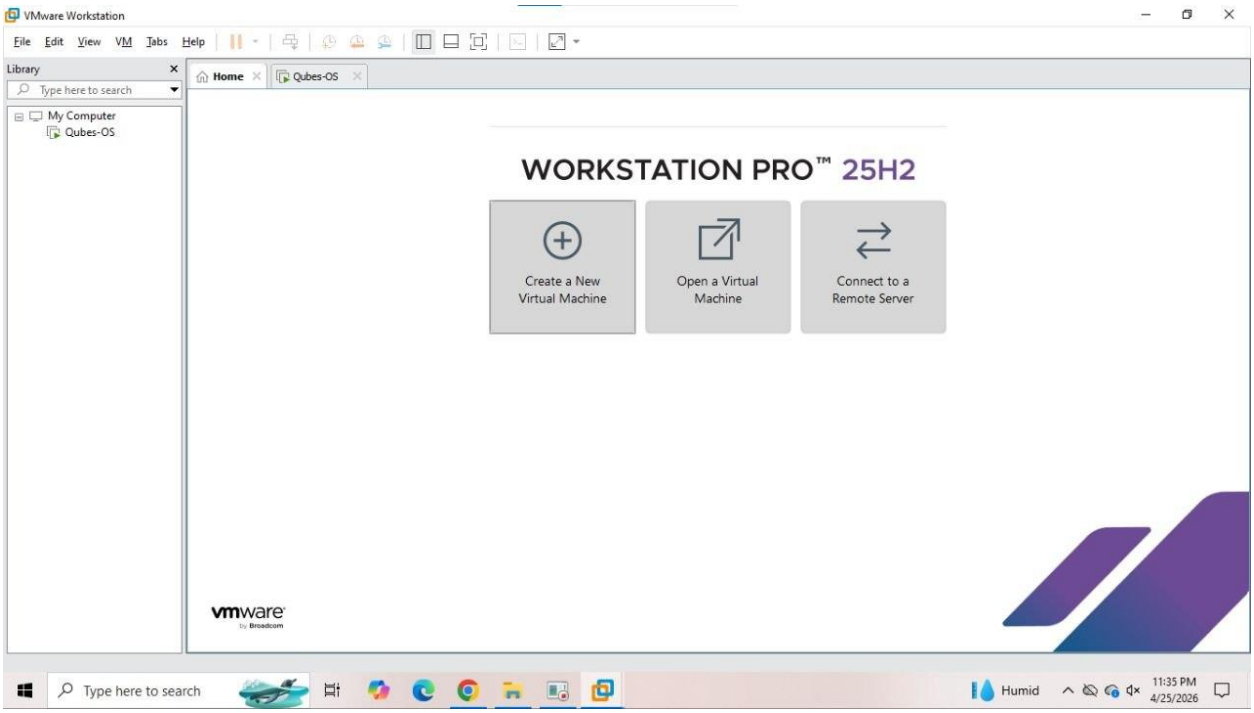

Fig 1: Creating a new virtual machine in VMware Workstation

2. Guest OS Configuration

- Select Other Linux (64-bit)

Explanation:
Qubes OS is based on a Linux kernel, but it is not listed as a standard option in VMware. Therefore, selecting “Other Linux (64-bit)” ensures compatibility and allows VMware to apply appropriate default configurations for the operating system.

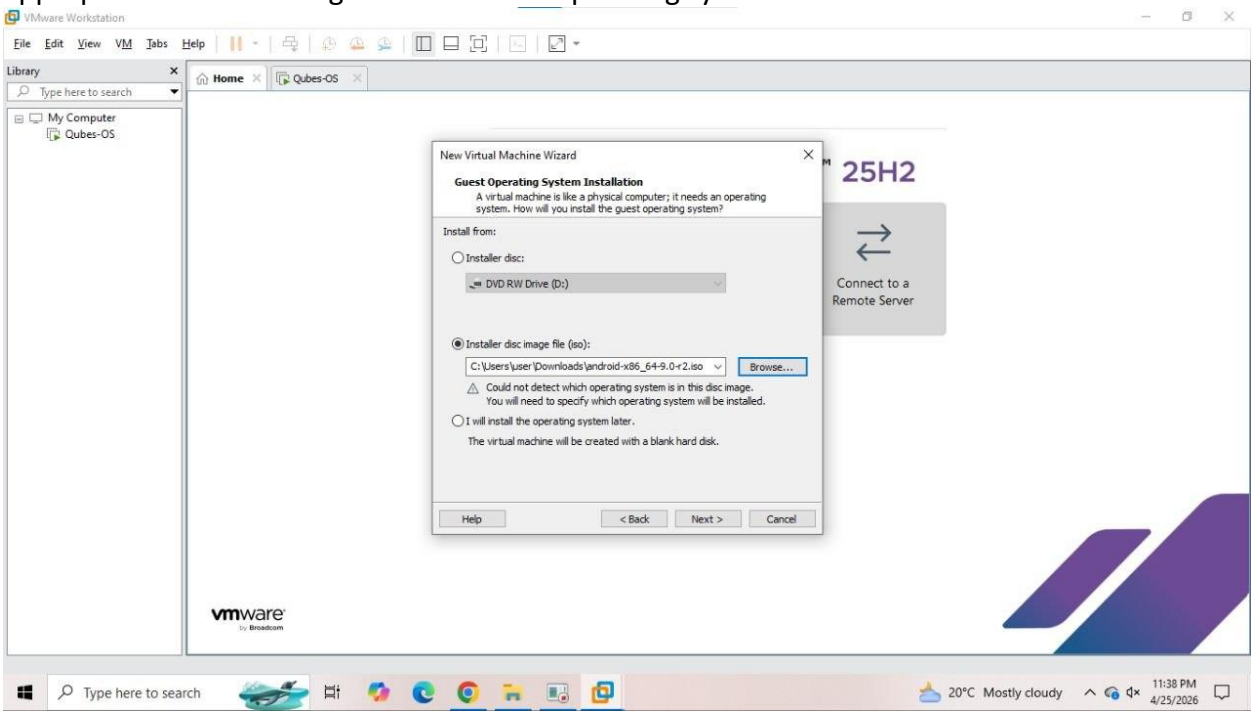

Fig 2: Guest operating system installation in VMware

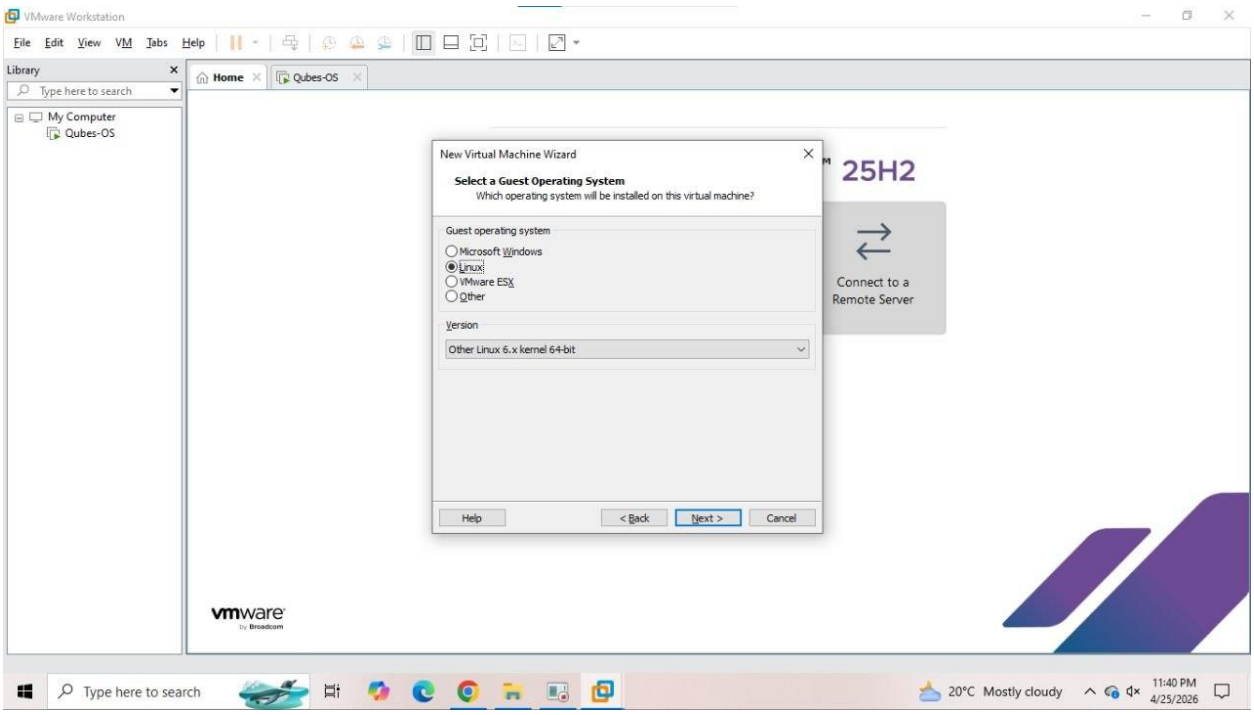

Fig 3: Selecting guest operating system type in VMware

3. Hardware Customization

- Allocate minimum 8 GB RAM
- Assign multiple CPU cores
- Enable:

  ❖ Virtualize Intel VT-x/EPT or AMD-V/RVI

➢ Without this step, Qubes OS will fail to start.

Explanation:
This is the most critical step in the installation process. Qubes OS relies heavily on virtualization because it runs multiple virtual machines internally using the Xen hypervisor.

- Allocating sufficient RAM ensures that multiple Qubes can run simultaneously.
- Assigning multiple CPU cores improves performance and allows parallel processing.
- Enabling virtualization extensions is essential for **nested virtualization**, allowing Qubes OS to run its own hypervisor inside VMware. Without this, the system cannot boot properly.

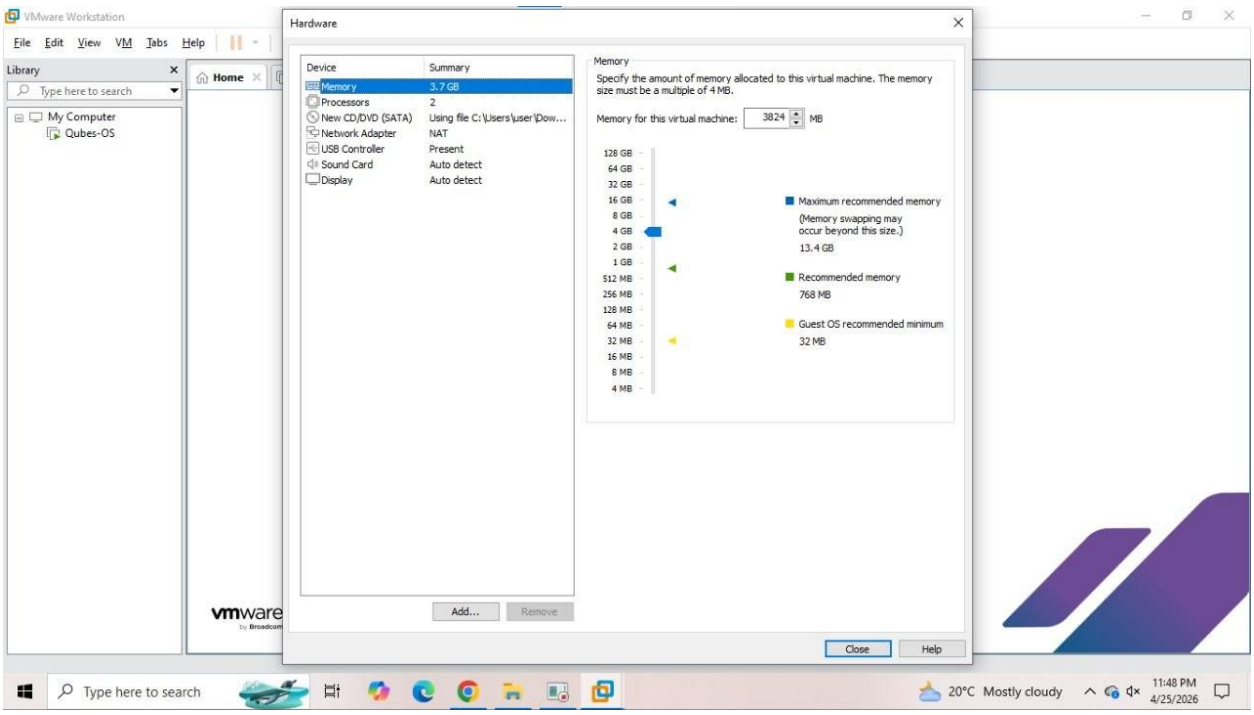

Fig 4: Allocating RAM and CPU resources to the virtual machine

4. Disk Configuration

- Allocate 70 GB disk space
- Select Store virtual disk as a single file

Explanation:
Qubes OS requires significant storage due to its use of multiple virtual machines. Allocating sufficient disk space ensures smooth operation.

Choosing “store as a single file” improves disk performance and is especially beneficial for LVM (Logical Volume Management), which Qubes OS uses to manage storage efficiently across multiple virtual environments.

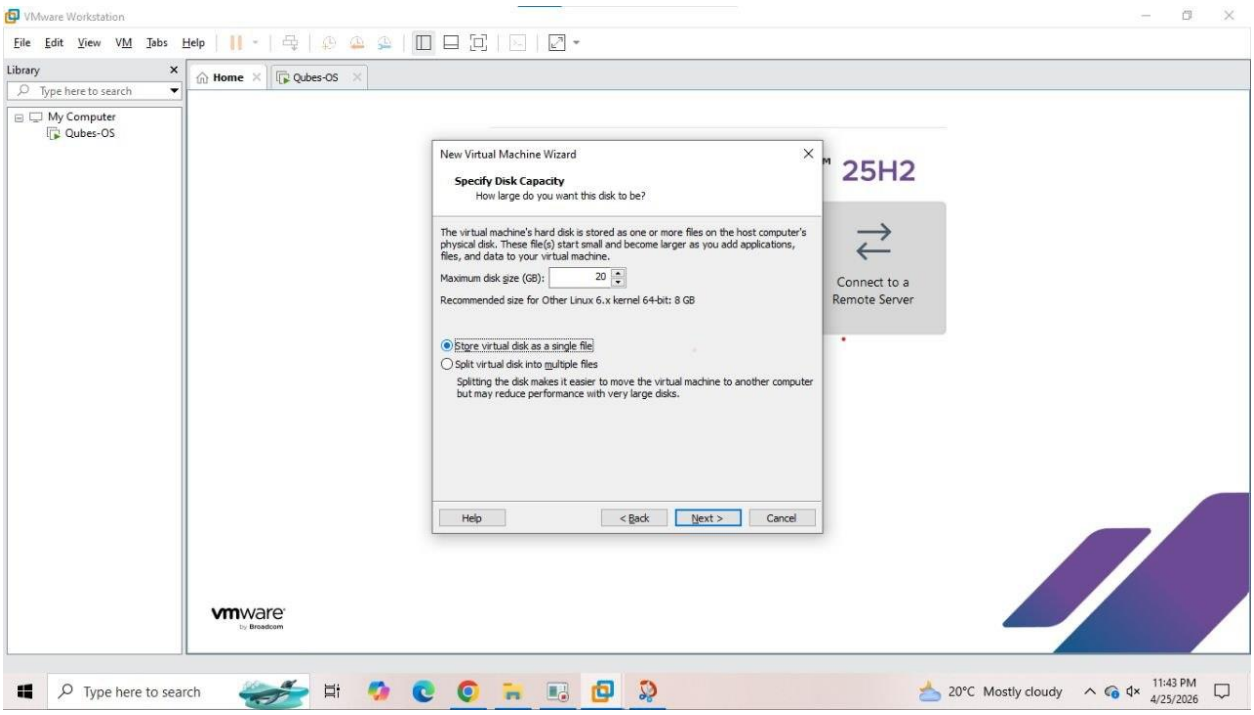

Fig 5: Configuring virtual disk size and storage format

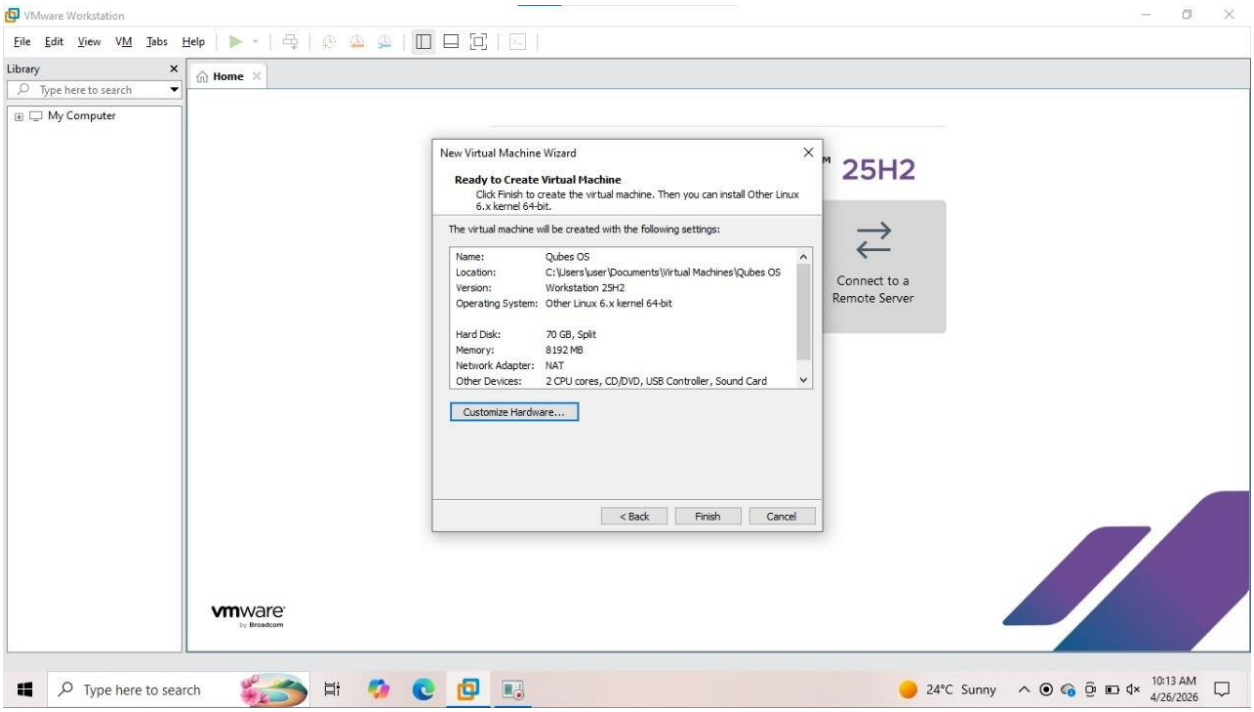

Fig 6:summary of specifications

5. Boot and Installation

- Start the VM
- Select Install Qubes OS
- Choose language and keyboard layout
- Select installation destination
- Use Automatic Partitioning

Explanation:
At this stage, the virtual machine boots from the ISO file and begins the installation process.

- Selecting language and keyboard ensures correct system localization.
- Choosing the installation destination defines where the OS will be installed.
- Automatic partitioning simplifies disk setup by allowing Qubes OS to configure partitions and LVM automatically, reducing the risk of manual errors.

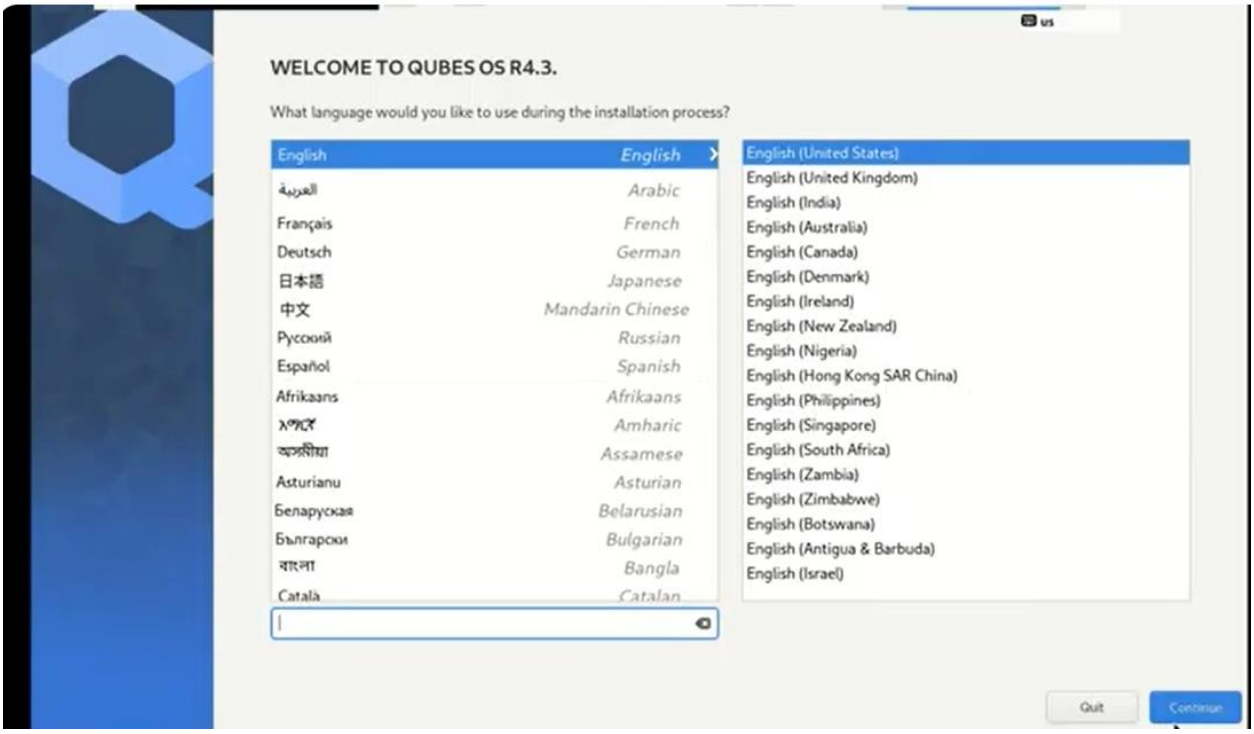

Fig 7: Selecting language and keyboard layout

6. User Setup

- Enter your **full name as username**
- Set a **secure password**

- **Explanation:**
  This step creates the primary user account for the system. Using my name to ensures proper identification.

Setting a strong password is important because Qubes OS is security-focused, and user credentials play a critical role in protecting system access.

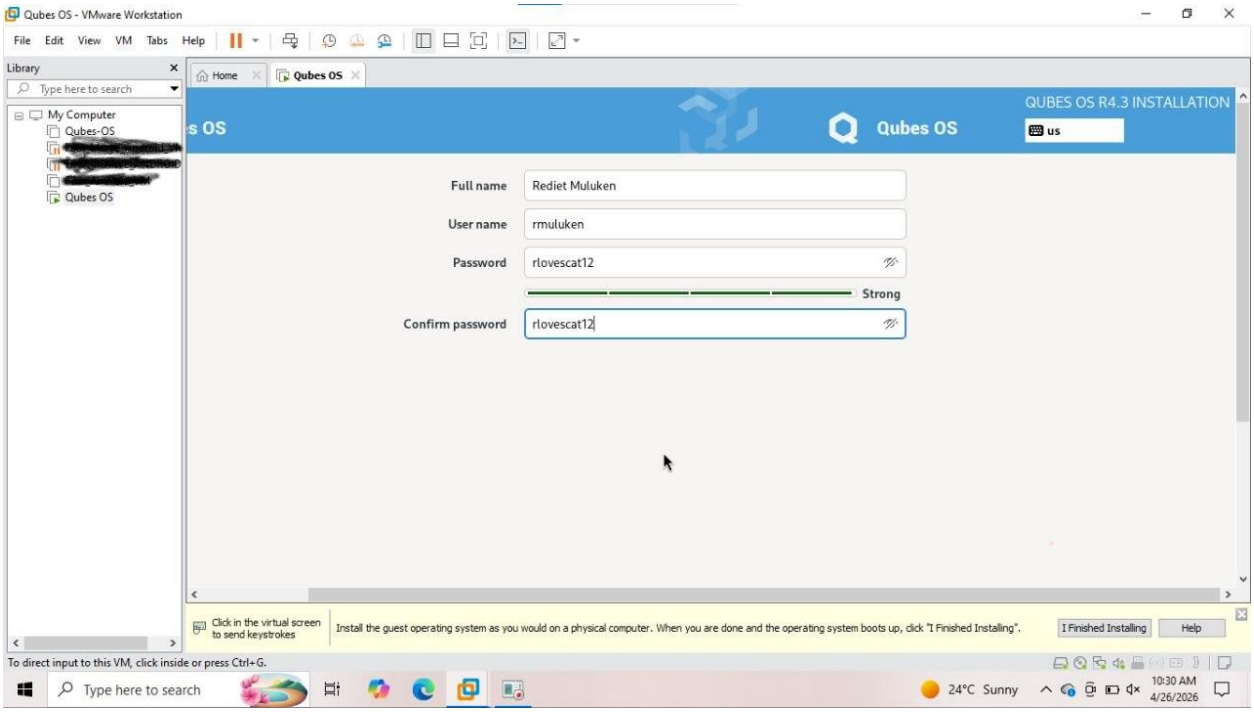

Fig 8: Creating user account and setting password

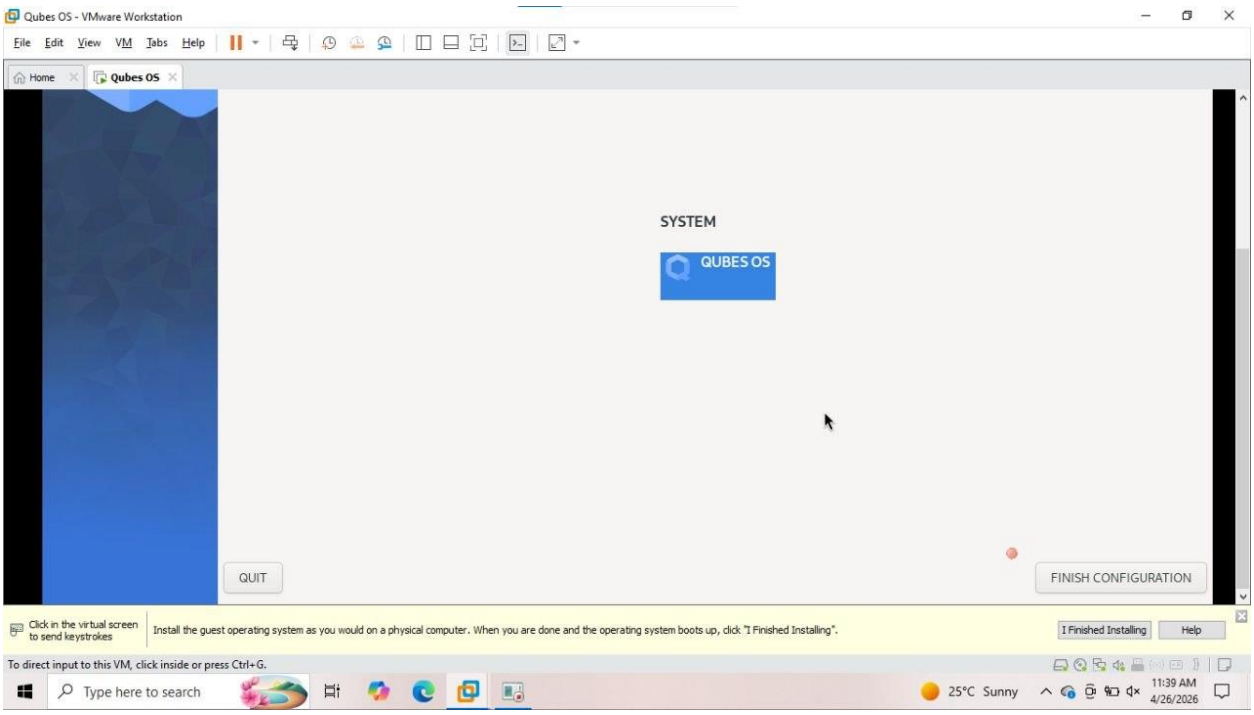

Fig 9: Finalizing system configuration to complete the Qubes OS installation

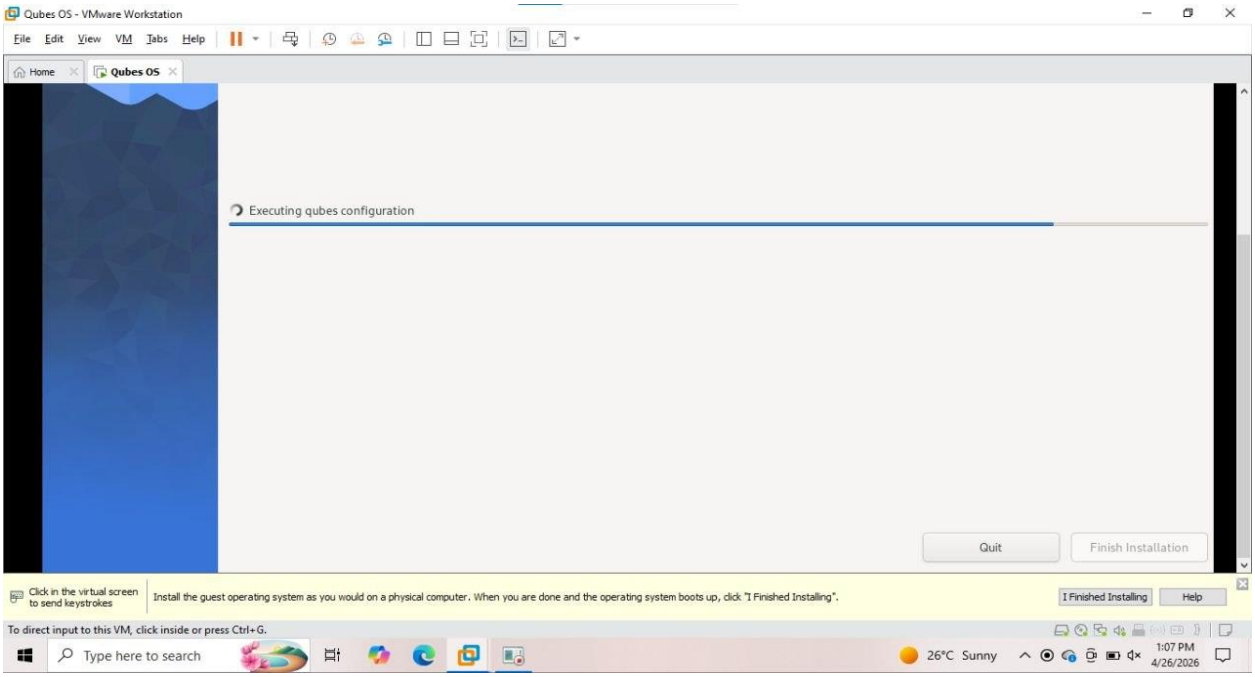

Fig 10: Configuration on process

7. Final Configuration

- After installation, reboot the system

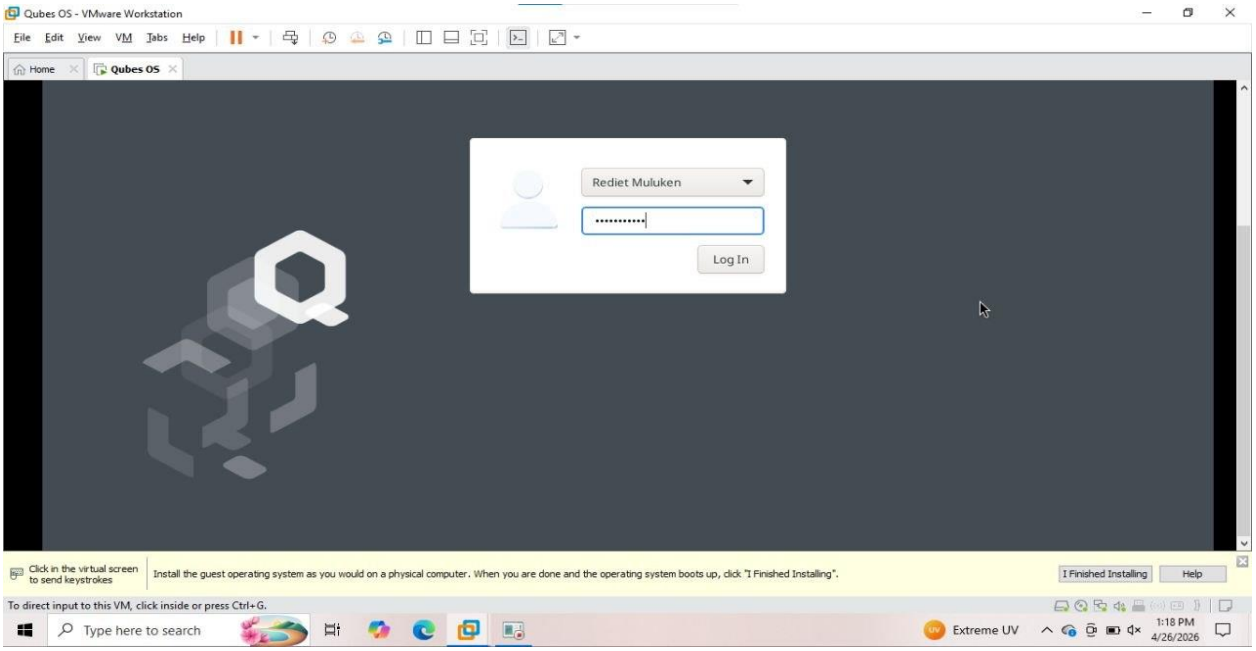

Fig 11: User login interface requiring authentication with username and password

8. Post-Installation Behavior

- System may be slow initially due to resource usage
- Qubes will initialize multiple VMs in the background

Explanation:
After installation, Qubes OS starts multiple background processes to initialize its virtual environments. This can temporarily reduce system performance.

This behavior is normal and expected because the system is setting up isolated domains. Once initialization is complete, performance stabilizes depending on available system resources.

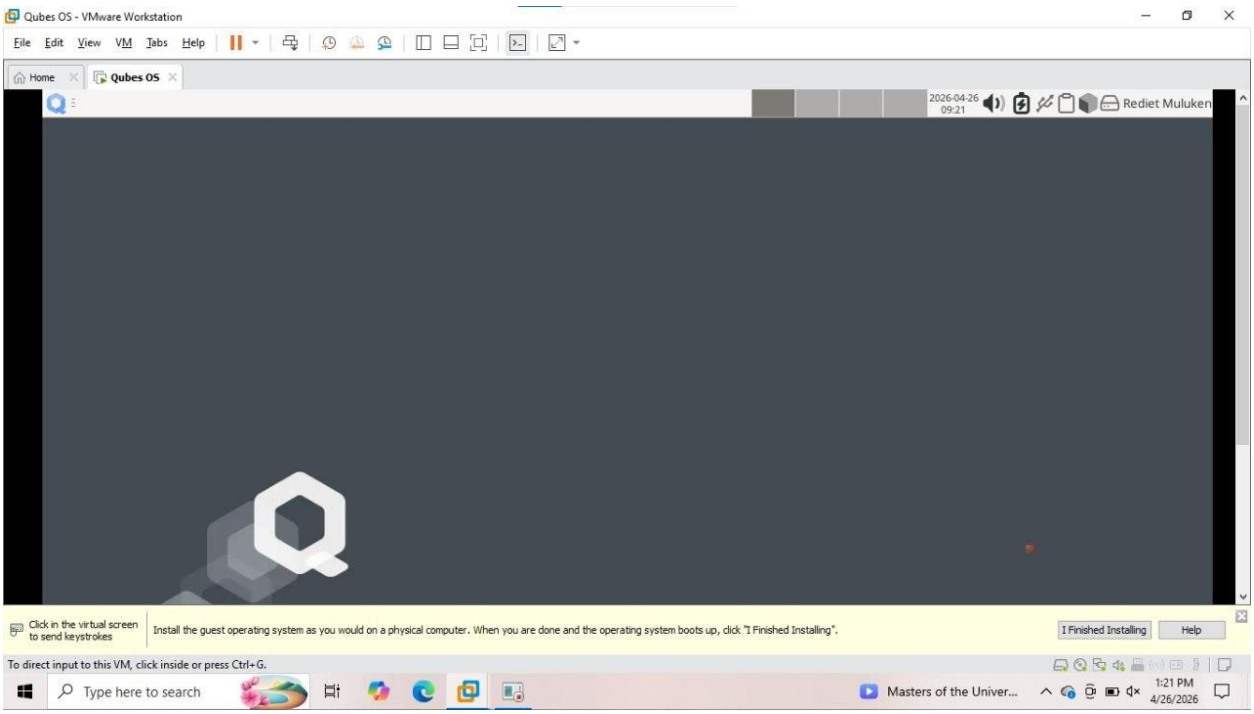

Fig 12: Qubes OS desktop interface after successful installation

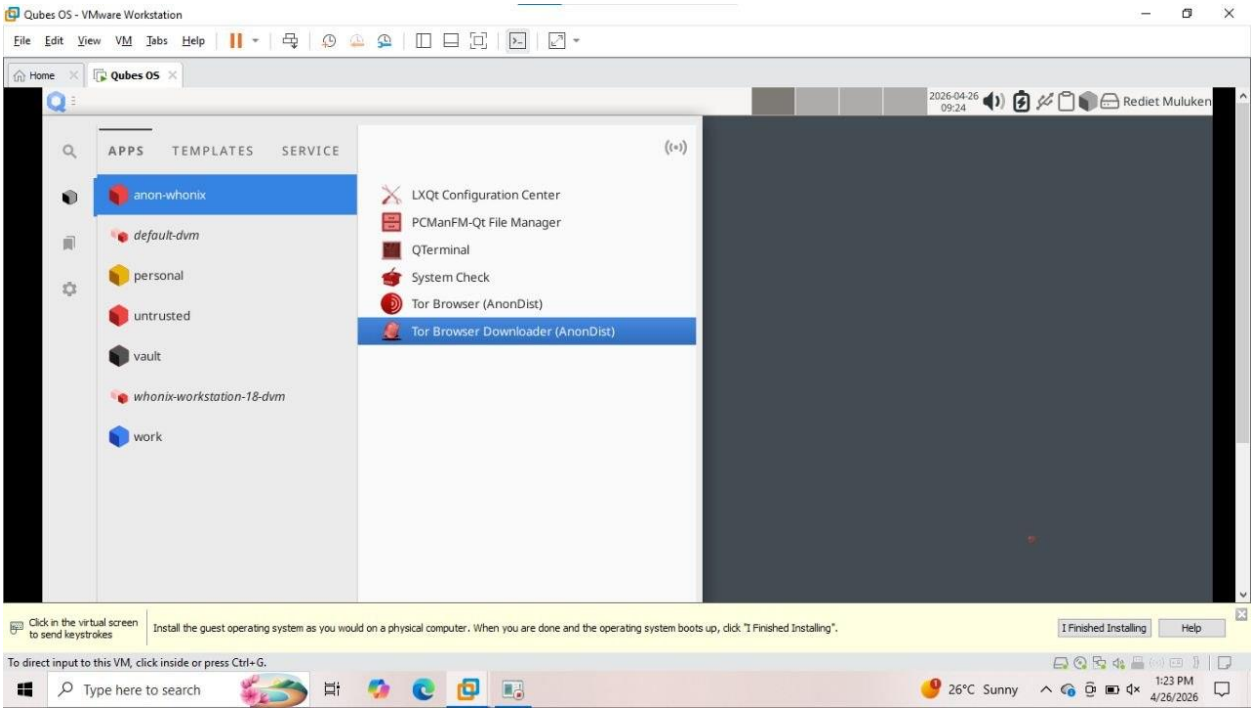

FIG 13: The Qubes os interface

E. Issues (Problems Faced)

During the installation of Qubes OS in VMware, several challenges were encountered. However, these issues were identified and resolved during the installation process, and the solutions have already been reflected in the installation steps described above.

The following are the key problems faced:

1. Nested Virtualization Requirement

Initially, Qubes OS could not boot properly because virtualization support was not fully enabled. This issue highlighted the importance of enabling virtualization features in VMware.

2. System Boot and Display Issues

At certain points, the system showed freezing behavior or delayed response during boot.
This was mainly due to limited resource allocation and virtualization overhead.

3. Bootloader (GRUB) Decision

During installation, the system prompted for GRUB installation, which required proper understanding to avoid boot failure.

4. Performance Limitations

After installation, the system appeared slow due to multiple Qubes initializing in the background.
This was expected behavior given the architecture of Qubes OS.

F. Solutions

All the issues mentioned above were resolved during the installation and configuration process, and the solutions are already incorporated into the installation steps.

1. Virtualization Configuration

Enabling:

- Virtualize Intel VT-x/EPT or AMD-V/RVI
  This step (included in hardware customization) resolved the boot issue.

2. Resource Allocation Adjustment

Allocating sufficient:

- RAM
- CPU cores

This improved system stability and prevented freezing during installation.

3. Correct Bootloader Selection

Selecting **"**Yes**"** for GRUB ensured that the system boots properly after installation.

4. Performance Optimization

Understanding that Qubes OS initializes multiple VMs helped interpret the initial slowness as normal behavior rather than a system error.

G. Filesystem Support

Qubes OS supports modern filesystems such as ext4 and XFS, both of which are widely used in Linux-based systems due to their reliability and performance.

Technical Explanation

Qubes OS utilizes Logical Volume Management (LVM) combined with thin provisioning. This approach allows flexible and efficient management of disk space. Instead of allocating fixed storage to each virtual machine, storage is dynamically assigned based on usage.

Each AppVM is based on a shared template. Instead of duplicating the entire filesystem, only the changes made by each VM are stored. This significantly reduces disk usage and improves efficiency.

Why These Filesystems?

- ext4**:** Known for its stability, speed, and widespread support. It is suitable for general-purpose usage.
- XFS**:** Designed for high performance and scalability, especially when handling large files and parallel operations.

These filesystems support important features such as:

- Fast read/write operations
- Snapshot capabilities
- Efficient storage allocation

This is essential in Qubes OS, where multiple virtual machines operate simultaneously and require efficient disk management.

H. Advantages and Disadvantages

Advantages

- **Strong Security Isolation:** Each application runs in its own isolated environment, preventing attacks from spreading across the system.
- **Secure Networking:** Network operations are handled in separate domains, reducing exposure to external threats.
- **Disposable Virtual Machines:** Users can create temporary environments that are automatically destroyed after use, ensuring no persistent malware remains.
- **Flexibility and Control:** Users can define trust levels and separate tasks based on sensitivity, improving overall system organization and security.

Disadvantages

- **High Resource Consumption:** Running multiple virtual machines requires significant CPU and memory resources, making it unsuitable for low-end systems.
- **Complexity:** The architecture of Qubes OS is more complex than traditional operating systems, which may be challenging for beginners to understand and use effectively.
- **Hardware Limitations:** Not all systems support the required virtualization features, limiting compatibility with older hardware.

Conclusion

This project provided a comprehensive understanding of operating system installation, virtualization, and security-focused system design. The successful installation of Qubes OS within a VMware Workstation environment demonstrated both theoretical knowledge and practical skills, particularly in configuring and managing nested virtualization.

Throughout the installation process, several challenges were encountered, including virtualization configuration issues, boot-related difficulties, and system performance limitations. These challenges were effectively resolved during the installation steps, reinforcing practical troubleshooting abilities and improving understanding of system requirements and behavior.

The project also highlighted the effectiveness of Qubes OS in implementing security through isolation and compartmentalization. By separating tasks into independent virtual machines, the system significantly reduces the risk of system-wide compromise and provides a strong security model compared to traditional operating systems.

In addition, this work enhanced skills in virtual machine configuration, hardware resource allocation, and system optimization. It also emphasized the importance of proper system planning, hardware compatibility, and performance considerations when working with advanced virtualization-based operating systems.

Overall, the objectives of the project were successfully achieved. The knowledge and experience gained from this work provide a strong foundation for understanding modern operating system design, virtualization technologies, and secure computing environments.

Future Outlook / Recommendations

The experience gained from this project provides valuable insights into both current limitations and future opportunities for improvement. Based on the knowledge acquired during the installation and analysis of Qubes OS, several recommendations can be made for future work and practical application.

1. Use of Dedicated Hardware

Although installing Qubes OS within VMware Workstation is useful for learning purposes, it is not ideal for real-world usage. This is mainly due to performance limitations and incomplete hardware access caused by nested virtualization.

For optimal performance and full utilization of system features, it is strongly recommended to install Qubes OS directly on compatible physical hardware. Running Qubes OS natively allows better interaction with the Xen hypervisor, improved graphics handling, and more efficient resource management. It also ensures that all security features function as intended without interference from an additional virtualization layer.

2. Increased System Resources

Qubes OS is a resource-intensive operating system because it runs multiple virtual machines simultaneously. Future implementations should prioritize systems with higher hardware specifications.

A minimum of 16 GB RAM is recommended, but higher capacities (such as 32 GB) would provide even better performance, especially when running multiple Qubes at the same time. Additionally, modern multi-core processors significantly improve virtualization efficiency by allowing better parallel execution of tasks.

Using high-speed SSD storage instead of traditional hard drives is also highly recommended, as it greatly enhances system responsiveness and reduces VM startup time.

3. Improved Understanding of Virtualization Technologies

This project highlights the importance of deeply understanding virtualization concepts. Future work should focus on studying hypervisors in more detail, particularly the Xen hypervisor used by Qubes OS.

A deeper understanding of virtualization layers, memory management, CPU scheduling, and hardware passthrough techniques can help in optimizing system performance and

troubleshooting complex issues. This knowledge is especially valuable in fields such as cloud computing, system administration, and infrastructure management.

4. Security-Oriented Practical Applications

Qubes OS offers a unique platform for applying security concepts in real-world scenarios. Future exploration should focus on using Qubes OS in practical security-focused tasks such as:

- **Secure Browsing:** Isolating web activities in separate Qubes to prevent tracking and malware infections
- **Malware Analysis:** Running suspicious files in isolated environments without risking the main system
- **Sensitive Data Handling:** Managing confidential information in highly secure, isolated domains

These applications allow users to apply theoretical knowledge of cybersecurity into real-life use cases, making Qubes OS highly relevant for professionals in security-related fields.

5. Automation and System Management

Future improvements can include the use of scripting and automation tools to manage Qubes more efficiently. For example, shell scripts can be used to automate the creation, configuration, and management of Qubes.

This not only improves system efficiency but also aligns with system programming concepts covered in the course. Automation reduces manual effort and minimizes the risk of configuration errors.

6. Performance Optimization and Monitoring

Another important recommendation is to explore performance monitoring and optimization techniques. Tools that monitor CPU usage, memory consumption, and VM activity can help identify system bottlenecks.

By analyzing system performance, users can make informed decisions about resource allocation and improve overall efficiency. This is especially important in environments where multiple virtual machines are running simultaneously.

7. Exploration of Advanced Qubes Features

Future work should also involve exploring advanced features of Qubes OS such as:

- Disposable VMs for temporary secure sessions

- Template-based VM management
- Secure file transfer between Qubes

Understanding and utilizing these features can significantly enhance both usability and security.

2. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a technology used in modern operating systems that allows a single physical computer to create and run multiple virtual environments. These environments behave like real independent computers but share the same physical hardware underneath.

Through virtualization, different operating systems (such as Linux, Windows, or Android-based systems) can run on one machine at the same time without interfering with each other. Each virtual environment is fully isolated, meaning it has its own memory, storage, and processing resources.

Why is Virtualization Used?

Virtualization is widely used in modern computing systems because it improves efficiency, security, and flexibility.

• **Security:** Virtualization isolates each system from others. If one virtual machine is attacked or infected with malware, the damage is contained and does not spread to the entire system.

• **Resource Efficiency:** Instead of using one computer for one operating system, virtualization allows multiple systems to run on a single machine. This reduces wasted hardware resources like CPU, memory, and storage.

• **Development and Testing:** Software developers can test applications in different environments without needing multiple physical computers. For example, they can test how an app behaves on different operating systems or configurations.

• **Cost Reduction:** Organizations save money by reducing the need for multiple physical machines. One powerful server can host many virtual systems, reducing hardware and maintenance costs.

• **Scalability and Flexibility:**Virtual machines can be created, copied, or deleted quickly, making it easy to scale systems based on demand.

How Does Virtualization Work?

Virtualization works through a special software layer called a hypervisor. The hypervisor sits between the physical hardware and the virtual machines and manages how resources are distributed.

It allocates CPU power, memory, storage, and network access to each virtual machine while ensuring they remain completely isolated from each other.

Each virtual machine behaves like a real computer with its own operating system and applications, even though they are all running on the same physical device.

Types of Hypervisors

• **Type 1 (Bare-Metal Hypervisor):** This type runs directly on the physical hardware without needing a host operating system. It provides high performance and strong security.
Examples include Xen and VMware ESXi.

• **Type 2 (Hosted Hypervisor):** This type runs on top of an existing operating system like Windows or Linux. It is easier to install and use but slightly less efficient.
Examples include VMware Workstation and VirtualBox.

Hardware Support for Virtualization

Modern processors include built-in hardware support for virtualization, which improves speed and efficiency.

• Intel VT-x (Intel Virtualization Technology)
• AMD-V (AMD Virtualization)

These technologies allow the hypervisor to run virtual machines more efficiently by reducing overhead and improving performance. Because of this, virtual machines today can perform almost as well as real physical systems.

Summary

In modern operating systems, virtualization plays a critical role by enabling multiple secure and efficient computing environments on a single machine. It improves resource usage, enhances security through isolation, reduces costs, and supports development and testing. The hypervisor and hardware virtualization technologies work together to make this possible.

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